Driver Scheduling and Management System (DSMS)

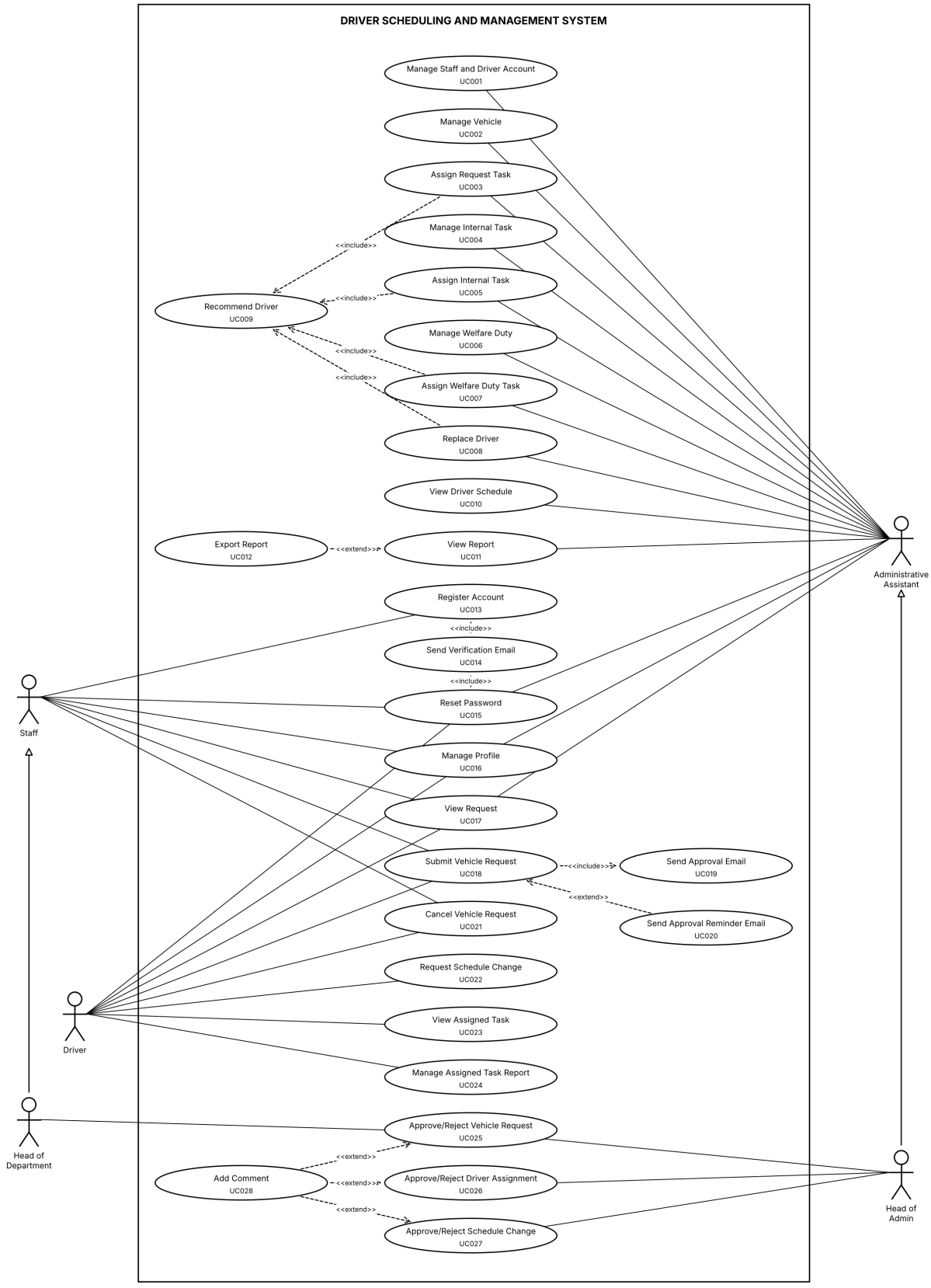
Use Case Diagram and Use Case Descriptions. Revised 14th November 2026.

Use Case List

ID	Use Case Name	Actor
UC001	Manage Staff and Driver Account	Administrative Assistant, Head of Admin
UC001.1	Create Driver Account	Administrative Assistant, Head of Admin
UC001.2	View Staff and Driver Account	Administrative Assistant, Head of Admin
UC001.3	Update Staff and Driver Account	Administrative Assistant, Head of Admin
UC001.4	Deactivate Staff and Driver Account	Administrative Assistant, Head of Admin
UC002	Manage Vehicle	Administrative Assistant, Head of Admin
UC002.1	Add Vehicle	Administrative Assistant, Head of Admin
UC002.2	View Vehicle	Administrative Assistant, Head of Admin
UC002.3	Update Vehicle	Administrative Assistant, Head of Admin
UC002.4	Deactivate Vehicle	Administrative Assistant, Head of Admin
UC003	Assign Request Task	Administrative Assistant, Head of Admin
UC004	Manage Internal Task	Administrative Assistant, Head of Admin
UC004.1	Add Internal Task	Administrative Assistant, Head of Admin
UC004.2	View Internal Task	Administrative Assistant, Head of Admin
UC004.3	Update Internal Task	Administrative Assistant, Head of Admin
UC004.4	Delete Internal Task	Administrative Assistant, Head of Admin
UC005	Assign Internal Task	Administrative Assistant, Head of Admin
UC006	Manage Welfare Duty	Administrative Assistant, Head of Admin
UC006.1	Create Welfare Duty Schedule	Administrative Assistant, Head of Admin
UC006.2	View Welfare Duty Schedule	Administrative Assistant, Head of Admin
UC006.3	Update Welfare Duty	Administrative Assistant, Head of Admin
UC006.4	Delete Welfare Duty	Administrative Assistant, Head of Admin
UC007	Assign Welfare Duty Task	Administrative Assistant, Head of Admin
UC008	Replace Driver	Administrative Assistant, Head of Admin
UC009	Recommend Driver	Administrative Assistant, Head of Admin
UC010	View Driver Schedule	Administrative Assistant, Head of Admin
UC011	View Report	Administrative Assistant, Head of Admin
UC012	Export Report	Administrative Assistant, Head of Admin
UC013	Register Account	Staff, Head of Department
UC014	Send Verification Email	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin
UC015	Reset Password	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin
UC016	Manage Profile	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin
UC016.1	View Profile	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin

ID	Use Case Name	Actor
UC016.2	Update Profile	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin
UC016.3	Change Password	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin
UC017	View Request	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin
UC018	Submit Vehicle Request	Staff, Head of Department, Driver
UC019	Send Approval Email	Staff, Head of Department, Driver
UC020	Send Approval Reminder Email	Head of Department
UC021	Cancel Vehicle Request	Staff, Head of Department, Driver
UC022	Request Schedule Change	Driver
UC023	View Assigned Task	Driver
UC024	Manage Assigned Task Report	Driver
UC024.1	Create Task Report	Driver
UC024.2	View Task Report	Driver
UC024.3	Update Task Report	Driver
UC025	Approve/Reject Vehicle Request	Head of Department, Head of Admin
UC026	Approve/Reject Driver Assignment	Head of Admin
UC027	Approve/Reject Schedule Change	Head of Admin
UC028	Add Comment	Head of Department, Head of Admin

Use Case Diagram



Use Case Descriptions

UC001 Manage Staff and Driver Account

Use Case ID	UC001
Use Case Name	Manage Staff and Driver Account
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to manage staff and driver accounts. This use case covers create driver account, view, update and deactivate staff and driver accounts, which are described in UC001.1, UC001.2, UC001.3 and UC001.4.
Preconditions	The Administrative Assistant is logged in.
Post Conditions	The account is created, displayed, updated or deactivated.
Priority	High
Frequency of Use	Medium
Normal Course of Events	Not applicable
Alternative Courses	Not applicable
Exceptions	Not applicable
Includes/Extends	Not applicable
Notes and Issues	• Staff and Head of Department register their own accounts through Register Account (UC013). Drivers do not register themselves. • A user account is never deleted. A deactivated user cannot log in, but all records of the user remain in the system.

UC001.1 Create Driver Account

Use Case ID	UC001.1
Use Case Name	Create Driver Account
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to create a new driver account. The system sends the login details to the driver's UIS email.
Preconditions	1. The Administrative Assistant is logged in. 2. The driver's staff ID and UIS email are not registered yet.
Post Conditions	The driver account is created with the Driver role and the login details are sent to the driver's UIS email.
Priority	High
Frequency of Use	Low

Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects User Account. / 1.1 System displays the list of users with their staff ID, name, department, role and status. 2. Administrative Assistant selects Create Driver. / 2.1 System displays the driver registration form. 3. Administrative Assistant enters the staff ID, name, UIS email, phone number, licence class, licence expiry date and start date of service, then selects Save. / 3.1 System checks that the staff ID and email are not registered. 3.2 System creates the account with the Driver role and a default password. 3.3 System sends the login details to the driver's UIS email and displays a success message.
Alternative Courses	Not applicable
Exceptions	3.1 The staff ID or email is already registered. The system displays a message that the account already exists. 3.2 A required field is empty. The system displays a message to fill in all required fields. 3.3 The email fails to send. The system keeps the account and displays a message that the email could not be sent.
Includes/Extends	Not applicable
Notes and Issues	• Changing the default password is optional for the driver. • The start date of service is used to count the driver's experience in Recommend Driver (UC009).

UC001.2 View Staff and Driver Account

Use Case ID	UC001.2
Use Case Name	View Staff and Driver Account
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to view the list and details of staff and driver accounts.
Preconditions	1. The Administrative Assistant is logged in. 2. At least one account exists in the system.
Post Conditions	The selected account details are displayed.
Priority	Medium
Frequency of Use	Medium
Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects User Account. / 1.1 System displays the list of users with their staff ID, name, department, role and status. 2. Administrative Assistant searches by staff ID or name, or filters by role. / 2.1 System displays the matching users. 3. Administrative Assistant selects a user. / 3.1 System displays the account details of the user. If the user is a driver, the system also displays the licence class, licence expiry date and start date of service.
Alternative Courses	Not applicable
Exceptions	2.1 No user matches the search. The system displays a message that no record is found.
Includes/Extends	Not applicable

Notes and Issues	Not applicable

UC001.3 Update Staff and Driver Account

Use Case ID	UC001.3
Use Case Name	Update Staff and Driver Account
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to update the department and role of a staff or driver account, and the licence details and start date of service of a driver account.
Preconditions	1. The Administrative Assistant is logged in. 2. The account exists in the system.
Post Conditions	The account is updated and the user is informed through email.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	

Actor	System
1. Administrative Assistant selects User Account.	1.1 System displays the list of users.
2. Administrative Assistant selects a user, then selects Edit.	2.1 System displays the account form with the current details. If the user is a driver, the form also displays the licence class, licence expiry date and start date of service.
3. Administrative Assistant updates the details, then selects Save.	3.1 System checks the details. 3.2 System updates the account. 3.3 System sends an email to the user about the change and displays a success message.

Alternative Courses	Not applicable
Exceptions	3.1 A required field is empty. The system displays a message to fill in all required fields. 3.2 The Administrative Assistant sets a user as Head of Department for a department that already has one. The system displays a message that the department already has a Head of Department.
Includes/Extends	Not applicable
Notes and Issues	Name, phone number, address and password are updated by the user through Manage Profile (UC016).

UC001.4 Deactivate Staff and Driver Account

Use Case ID	UC001.4
Use Case Name	Deactivate Staff and Driver Account
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to deactivate a staff or driver account so the user can no longer log in.
Preconditions	1. The Administrative Assistant is logged in. 2. The account exists and is active.
Post Conditions	The account is deactivated and the user cannot log in.
Priority	Medium
Frequency of Use	Low

Normal Course of Events	<table><tr><th>Actor</th><th>System</th></tr><tr><td>1. Administrative Assistant selects User Account.</td><td>1.1 System displays the list of users.</td></tr><tr><td>2. Administrative Assistant selects an active user, then selects Deactivate.</td><td>2.1 System displays a confirmation pop-up.</td></tr><tr><td>3. Administrative Assistant selects Confirm.</td><td>3.1 System deactivates the account and displays a success message.</td></tr></table>
Alternative Courses	Not applicable
Exceptions	3.1 The user is a driver with upcoming tasks. The system displays the tasks and asks the Administrative Assistant to replace the driver first (UC008). 3.2 The user is a Head of Department with pending requests. The system displays a message that the pending requests must be handled first.
Includes/Extends	Not applicable
Notes and Issues	A deactivated user cannot log in, but all records of the user remain in the system.

UC002 Manage Vehicle

Use Case ID	UC002
Use Case Name	Manage Vehicle
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to manage vehicle records. This use case covers add, view, update and deactivate vehicle, which are described in UC002.1, UC002.2, UC002.3 and UC002.4.
Preconditions	The Administrative Assistant is logged in.
Post Conditions	The vehicle is added, displayed, updated or deactivated.
Priority	High
Frequency of Use	Medium
Normal Course of Events	Not applicable
Alternative Courses	Not applicable
Exceptions	Not applicable
Includes/Extends	Not applicable
Notes and Issues	• The vehicle types are car, MPV, van, bus and lorry. • A vehicle is never deleted. A deactivated vehicle is not shown when a request is submitted.

UC002.1 Add Vehicle

Use Case ID	UC002.1
Use Case Name	Add Vehicle
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to add a new vehicle to the system.
Preconditions	1. The Administrative Assistant is logged in. 2. The plate number is not registered yet.
Post Conditions	The vehicle is added with the Active status.
Priority	High
Frequency of Use	Low

Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects Vehicle. / 1.1 System displays the list of vehicles with their plate number, vehicle type, seating capacity and status. 2. Administrative Assistant selects Add Vehicle. / 2.1 System displays the add vehicle form. 3. Administrative Assistant enters the plate number, vehicle type, brand, model, fuel type, seating capacity, required licence class, current mileage and current fuel level, then selects Save. / 3.1 System checks that the plate number is not registered. 3.2 System adds the vehicle with the Active status and displays a success message.
Alternative Courses	Not applicable
Exceptions	3.1 The plate number is already registered. The system displays a message that the vehicle already exists. 3.2 A required field is empty. The system displays a message to fill in all required fields.
Includes/Extends	Not applicable
Notes and Issues	• The fuel level options are Full, 3/4, 1/2, 1/4 and Empty. • The required licence class is used to filter drivers in Recommend Driver (UC009).

UC002.2 View Vehicle

Use Case ID	UC002.2
Use Case Name	View Vehicle
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to view the list and details of vehicles.
Preconditions	1. The Administrative Assistant is logged in. 2. At least one vehicle exists in the system.
Post Conditions	The selected vehicle details are displayed.
Priority	Medium
Frequency of Use	High
Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects Vehicle. / 1.1 System displays the list of vehicles with their plate number, vehicle type, seating capacity and status. 2. Administrative Assistant filters the list by vehicle type or status. / 2.1 System displays the matching vehicles. 3. Administrative Assistant selects a vehicle. / 3.1 System displays the vehicle details, current mileage, current fuel level and upcoming bookings.
Alternative Courses	Not applicable
Exceptions	2.1 No vehicle matches the filter. The system displays a message that no record is found.
Includes/Extends	Not applicable
Notes and Issues	Not applicable

UC002.3 Update Vehicle

Use Case ID	UC002.3

Use Case Name	Update Vehicle
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to update the current fuel level, current mileage and status of a vehicle.
Preconditions	1. The Administrative Assistant is logged in. 2. The vehicle exists and is active.
Post Conditions	The vehicle details are updated.
Priority	Medium
Frequency of Use	Medium
Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects Vehicle. / 1.1 System displays the list of vehicles. 2. Administrative Assistant selects a vehicle, then selects Edit. / 2.1 System displays the current fuel level, current mileage and status. 3. Administrative Assistant updates the details, then selects Save. / 3.1 System checks the details. 3.2 System updates the vehicle and displays a success message.
Alternative Courses	Not applicable
Exceptions	3.1 The new mileage is lower than the current mileage. The system displays a message that the mileage is not valid. 3.2 The status is changed to Under Maintenance while the vehicle has upcoming bookings. The system displays the affected bookings and asks the Administrative Assistant to assign another vehicle to each booking.
Includes/Extends	Not applicable
Notes and Issues	• Only the current fuel level, current mileage and status can be updated. • The status options are Active and Under Maintenance.

UC002.4 Deactivate Vehicle

Use Case ID	UC002.4
Use Case Name	Deactivate Vehicle
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to deactivate a vehicle that is no longer used.
Preconditions	1. The Administrative Assistant is logged in. 2. The vehicle exists and is active.
Post Conditions	The vehicle status is changed to Inactive.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects Vehicle. / 1.1 System displays the list of vehicles. 2. Administrative Assistant selects an active vehicle, then selects Deactivate. / 2.1 System displays a confirmation pop-up. 3. Administrative Assistant selects Confirm. / 3.1 System deactivates the vehicle and displays a success message.
Alternative Courses	Not applicable
Exceptions	3.1 The vehicle has upcoming bookings. The system displays the affected bookings and asks the Administrative Assistant to assign another vehicle to each booking.

Includes/Extends	Not applicable
Notes and Issues	A deactivated vehicle is not shown when a request is submitted.

UC003 Assign Request Task

Use Case ID	UC003
Use Case Name	Assign Request Task
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to assign a vehicle and a driver to a request approved by the Head of Department. The system recommends suitable drivers.
Preconditions	1. The Administrative Assistant is logged in. 2. The request status is Approved by Head of Department.
Post Conditions	1. The vehicle and driver are assigned. 2. The request is sent to the Head of Admin for approval.
Priority	High
Frequency of Use	High
Normal Course of Events	See table below
Alternative Courses	A1 Same driver for the return trip. At step 4, the Administrative Assistant selects Same Driver for Return Trip. The system uses the outbound driver for the return trip. A2 Driver outside the recommendation. At step 4, the Administrative Assistant chooses a driver who is not in the recommended list. The system displays a warning with the driver's task count this month. The Administrative Assistant selects Confirm. A3 Vendor bus. At step 3, not enough buses are available. The Administrative Assistant selects Vendor Bus and enters the vendor name and bus plate number. The system records the vendor bus for the request.
Exceptions	2.2 The distance entered by the requester is not correct. The Administrative Assistant corrects the distance and the system calculates the charge again. 3.1 No driver is available for the period. The system displays a message that no driver is available.
Includes/Extends	<<include>> UC009 Recommend Driver
Notes and Issues	• The Administrative Assistant selects the specific vehicle from the vehicles reserved when the request was submitted. • Pay and Claim and Self Drive requests are not assigned in this use case because no UIS driver is needed.

Normal Course of Events

Actor	System
1. Administrative Assistant selects Assign Request Task.	1.1 System displays the requests waiting for driver assignment, with Urgent requests at the top.
2. Administrative Assistant selects a request.	2.1 System displays the request details and the vehicles reserved for the request. 2.2 If the request is a Personal request, the system displays the round-trip distance entered by the requester and calculates the charge at RM3 per kilometre.
3. Administrative Assistant selects the vehicle.	3.1 System displays the recommended drivers for the vehicle (UC009).
4. Administrative Assistant selects a driver for the outbound trip and the return trip.	4.1 System records the driver for the vehicle.
5. Administrative Assistant repeats steps 3 and 4 for each vehicle in the request, then selects Submit.	5.1 System saves the assignment and changes the status to Driver Assigned. 5.2 System sends the request to the Head of Admin for approval (UC026).

UC004 Manage Internal Task

Use Case ID	UC004
Use Case Name	Manage Internal Task
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to manage internal tasks such as workshop runs, refuelling and test drives. This use case covers add, view, update and delete internal task, which are described in UC004.1, UC004.2, UC004.3 and UC004.4.
Preconditions	The Administrative Assistant is logged in.
Post Conditions	The internal task is added, displayed, updated or deleted.
Priority	Medium
Frequency of Use	Medium
Normal Course of Events	Not applicable
Alternative Courses	Not applicable
Exceptions	Not applicable
Includes/Extends	Not applicable
Notes and Issues	• The task types are Workshop, Refuelling and Test Drive. • The driver is assigned through Assign Internal Task (UC005).

UC004.1 Add Internal Task

Use Case ID	UC004.1
Use Case Name	Add Internal Task
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to add a new internal task and reserve a vehicle for it.
Preconditions	The Administrative Assistant is logged in.
Post Conditions	1. The internal task is saved with the Waiting for Driver status. 2. The vehicle is reserved for the task period.
Priority	Medium
Frequency of Use	Medium
Normal Course of Events	

Actor	**System**
1. Administrative Assistant selects Internal Task.	1.1 System displays the internal tasks with their date, task type, vehicle, driver and status.
2. Administrative Assistant selects Add Task.	2.1 System displays the internal task form.
3. Administrative Assistant selects the task type, enters the date, start time, end time and description, selects the vehicle, then selects Save.	3.1 System checks that the vehicle is available for that period. 3.2 System saves the task with the Waiting for Driver status. 3.3 System reserves the vehicle for that period and displays a success message.

Alternative Courses	Not applicable

Exceptions	3.1 The vehicle is not available for that period. The system displays a message that the vehicle is not available. 3.2 A required field is empty. The system displays a message to fill in all required fields.
Includes/Extends	Not applicable
Notes and Issues	Not applicable

UC004.2 View Internal Task

Use Case ID	UC004.2
Use Case Name	View Internal Task
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to view the list and details of internal tasks.
Preconditions	1. The Administrative Assistant is logged in. 2. At least one internal task exists in the system.
Post Conditions	The selected internal task details are displayed.
Priority	Medium
Frequency of Use	Medium
Normal Course of Events	

Actor	System
1. Administrative Assistant selects Internal Task.	1.1 System displays the internal tasks with their date, task type, vehicle, driver and status.
2. Administrative Assistant filters the list by date, task type or status.	2.1 System displays the matching tasks.
3. Administrative Assistant selects a task.	3.1 System displays the task details.

Alternative Courses	Not applicable
Exceptions	2.1 No task matches the filter. The system displays a message that no record is found.
Includes/Extends	Not applicable
Notes and Issues	Not applicable

UC004.3 Update Internal Task

Use Case ID	UC004.3
Use Case Name	Update Internal Task
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to update the details of an internal task.
Preconditions	1. The Administrative Assistant is logged in. 2. The internal task exists and is not completed.
Post Conditions	The internal task is updated and the assigned driver is notified, if any.
Priority	Medium
Frequency of Use	Low

Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects Internal Task. / 1.1 System displays the internal tasks. 2. Administrative Assistant selects a task, then selects Edit. / 2.1 System displays the task form with the current details. 3. Administrative Assistant updates the details, then selects Save. / 3.1 System checks that the vehicle is available for the new period. 3.2 System updates the task and the vehicle reservation. 3.3 System sends a notification to the assigned driver, if any, and displays a success message.
Alternative Courses	Not applicable
Exceptions	3.1 The vehicle is not available for the new period. The system displays a message that the vehicle is not available. 3.2 A required field is empty. The system displays a message to fill in all required fields.
Includes/Extends	Not applicable
Notes and Issues	Not applicable

UC004.4 Delete Internal Task

Use Case ID	UC004.4
Use Case Name	Delete Internal Task
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to delete an internal task that is no longer needed.
Preconditions	1. The Administrative Assistant is logged in. 2. The internal task exists and is not completed.
Post Conditions	The internal task is deleted, the vehicle reservation is released and the assigned driver is notified, if any.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects Internal Task. / 1.1 System displays the internal tasks. 2. Administrative Assistant selects a task, then selects Delete. / 2.1 System displays a confirmation pop-up. 3. Administrative Assistant selects Confirm. / 3.1 System deletes the task and releases the vehicle reservation. 3.2 System sends a notification to the assigned driver, if any, and displays a success message.
Alternative Courses	Not applicable
Exceptions	Not applicable
Includes/Extends	Not applicable
Notes and Issues	A completed internal task cannot be deleted because it is counted in the driver's task count.

UC005 Assign Internal Task

Use Case ID	UC005
Use Case Name	Assign Internal Task
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to assign a driver to an internal task. The system recommends suitable drivers.
Preconditions	1. The Administrative Assistant is logged in. 2. The internal task status is Waiting for Driver.
Post Conditions	1. The driver is assigned. 2. The task is sent to the Head of Admin for approval.
Priority	Medium
Frequency of Use	Medium
Normal Course of Events	

Actor	System
1. Administrative Assistant selects Internal Task, then selects a task with the Waiting for Driver status.	1.1 System displays the task details.
2. Administrative Assistant selects Assign Driver.	2.1 System displays the recommended drivers for the vehicle (UC009).
3. Administrative Assistant selects a driver, then selects Submit.	3.1 System saves the assignment and changes the status to Driver Assigned. 3.2 System sends the task to the Head of Admin for approval (UC026).

Alternative Courses	A1 Driver outside the recommendation. At step 3, the Administrative Assistant chooses a driver who is not in the recommended list. The system displays a warning with the driver's task count this month. The Administrative Assistant selects Confirm.
Exceptions	2.1 No driver is available for the period. The system displays a message that no driver is available.
Includes/Extends	<<include>> UC009 Recommend Driver
Notes and Issues	Internal tasks are counted in each driver's task count for driver recommendation.

UC006 Manage Welfare Duty

Use Case ID	UC006
Use Case Name	Manage Welfare Duty
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to manage the weekly welfare duty schedule. Welfare duty is a 24-hour standby duty using a car or van. This use case covers create, view, update and delete welfare duty, which are described in UC006.1, UC006.2, UC006.3 and UC006.4.
Preconditions	The Administrative Assistant is logged in.
Post Conditions	The welfare duty schedule is created, displayed, updated or deleted.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	Not applicable
Alternative Courses	Not applicable
Exceptions	Not applicable
Includes/Extends	Not applicable

Notes and Issues	• Only cars and vans can be selected for welfare duty. • The drivers are assigned through Assign Welfare Duty Task (UC007).

UC006.1 Create Welfare Duty Schedule

Use Case ID	UC006.1
Use Case Name	Create Welfare Duty Schedule
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to create the welfare duty schedule for a week by selecting a vehicle for each day.
Preconditions	1. The Administrative Assistant is logged in. 2. No schedule exists yet for the selected week.
Post Conditions	1. The schedule is saved with the Waiting for Driver status. 2. The vehicle is reserved for each 24-hour duty.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	

Actor	System
1. Administrative Assistant selects Welfare Duty.	1.1 System displays the welfare duty schedule by week.
2. Administrative Assistant selects Create Schedule, then selects the week.	2.1 System displays the seven days of the week.
3. Administrative Assistant selects the vehicle for each day, then selects Save.	3.1 System checks that each vehicle is available on that day. 3.2 System saves the schedule with the Waiting for Driver status. 3.3 System reserves the vehicle for each 24-hour duty and displays a success message.

Alternative Courses	Not applicable
Exceptions	2.1 A schedule already exists for the selected week. The system displays a message that the schedule already exists. 3.1 The vehicle is not available on that day. The system displays a message that the vehicle is not available.
Includes/Extends	Not applicable
Notes and Issues	Not applicable

UC006.2 View Welfare Duty Schedule

Use Case ID	UC006.2
Use Case Name	View Welfare Duty Schedule
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to view the welfare duty schedule by week.
Preconditions	1. The Administrative Assistant is logged in. 2. At least one welfare duty schedule exists.
Post Conditions	The schedule for the selected week is displayed.
Priority	Medium
Frequency of Use	Medium

Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects Welfare Duty. / 1.1 System displays the welfare duty schedule by week. 2. Administrative Assistant selects a week. / 2.1 System displays the vehicle, driver and status for each day of that week.
Alternative Courses	Not applicable
Exceptions	2.1 No schedule exists for the selected week. The system displays a message that no record is found.
Includes/Extends	Not applicable
Notes and Issues	Not applicable

UC006.3 Update Welfare Duty

Use Case ID	UC006.3
Use Case Name	Update Welfare Duty
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to change the vehicle for a day in the welfare duty schedule.
Preconditions	1. The Administrative Assistant is logged in. 2. The welfare duty for that day exists and has not passed.
Post Conditions	The duty is updated and the assigned driver is notified, if any.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects Welfare Duty, then selects a week. / 1.1 System displays the seven days of the week. 2. Administrative Assistant selects a day, then selects Edit. / 2.1 System displays the current vehicle for that day. 3. Administrative Assistant changes the vehicle, then selects Save. / 3.1 System checks that the vehicle is available on that day. 3.2 System updates the duty and the vehicle reservation. 3.3 System sends a notification to the assigned driver, if any, and displays a success message.
Alternative Courses	Not applicable
Exceptions	3.1 The vehicle is not available on that day. The system displays a message that the vehicle is not available.
Includes/Extends	Not applicable
Notes and Issues	Not applicable

UC006.4 Delete Welfare Duty

Use Case ID	UC006.4
Use Case Name	Delete Welfare Duty
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to delete the welfare duty for a day.

Preconditions	1. The Administrative Assistant is logged in. 2. The welfare duty for that day exists and has not passed.
Post Conditions	The duty is deleted, the vehicle reservation is released and the assigned driver is notified, if any.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects Welfare Duty, then selects a week. / 1.1 System displays the seven days of the week. 2. Administrative Assistant selects a day, then selects Delete. / 2.1 System displays a confirmation pop-up. 3. Administrative Assistant selects Confirm. / 3.1 System deletes the duty and releases the vehicle reservation. 3.2 System sends a notification to the assigned driver, if any, and displays a success message.
Alternative Courses	Not applicable
Exceptions	Not applicable
Includes/Extends	Not applicable
Notes and Issues	A past welfare duty cannot be deleted because it is counted in the driver's task count.

UC007 Assign Welfare Duty Task

Use Case ID	UC007
Use Case Name	Assign Welfare Duty Task
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to assign a driver to each day of the weekly welfare duty schedule. The system recommends suitable drivers.
Preconditions	1. The Administrative Assistant is logged in. 2. The welfare duty schedule for the week exists.
Post Conditions	1. The drivers are assigned. 2. The schedule is sent to the Head of Admin for approval.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects Welfare Duty, then selects a week with the Waiting for Driver status. / 1.1 System displays the seven days of the week. 2. Administrative Assistant selects a day. / 2.1 System displays the recommended drivers for that day (UC009). 3. Administrative Assistant selects a driver, repeats steps 2 and 3 for each day, then selects Submit. / 3.1 System saves the assignments and changes the status to Driver Assigned. 3.2 System sends the schedule to the Head of Admin for approval (UC026).
Alternative Courses	A1 Driver outside the recommendation. At step 3, the Administrative Assistant chooses a driver who is not in the recommended list. The system displays a warning with the driver's task count this month. The Administrative Assistant selects Confirm.

Exceptions	2.1 No driver is available on that day. The system displays a message that no driver is available. 3.1 Not every day has a driver when Submit is selected. The system displays a message that every day needs a driver.
Includes/Extends	<<include>> UC009 Recommend Driver
Notes and Issues	• Welfare duty is counted in each driver's task count, so the duty is rotated among drivers. • Drivers with an approved schedule change on that day are not recommended.

UC008 Replace Driver

Use Case ID	UC008
Use Case Name	Replace Driver
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant to replace the driver of an approved task when the driver cannot do the task. The reason for every replacement is recorded.
Preconditions	1. The Administrative Assistant is logged in. 2. The task status is Approved and the task has not started.
Post Conditions	1. The new driver is sent to the Head of Admin for approval. 2. The replacement is recorded with the reason.
Priority	High
Frequency of Use	Medium
Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects Schedule, then selects a task. / 1.1 System displays the task details and the assigned driver. 2. Administrative Assistant selects Replace Driver. / 2.1 System displays the recommended drivers for the vehicle (UC009) and a field for the reason. 3. Administrative Assistant selects a new driver, enters the reason, then selects Confirm. / 3.1 System records the replacement with the reason. 3.2 System sends the replacement to the Head of Admin for approval (UC026).
Alternative Courses	A1 Replacement from another use case. At step 1, the replacement comes from an approved schedule change (UC027) or a driver account being deactivated (UC001.4). The system opens the affected task directly. The flow continues from step 2. A2 Driver outside the recommendation. At step 3, the Administrative Assistant chooses a driver who is not in the recommended list. The system displays a warning with the driver's task count this month. The Administrative Assistant selects Confirm.
Exceptions	2.1 No driver is available for the period. The system displays a message that no driver is available. 3.1 The reason is empty. The system displays a message that the reason is required.
Includes/Extends	<<include>> UC009 Recommend Driver
Notes and Issues	• Every replacement is recorded with the previous driver, new driver, reason, date and the user who made the change. These records are shown in View Report (UC011). • The replacement does not go to the Head of Department. Once the Head of Admin approves it, the system notifies the previous driver, the new driver and the requester.

UC009 Recommend Driver

Use Case ID	UC009
Use Case Name	Recommend Driver

Actor	Administrative Assistant, Head of Admin
Description	Recommends suitable drivers for a task. The system filters drivers by licence class and availability, then ranks them with a weighted score.
Preconditions	Assign Request Task (UC003), Assign Internal Task (UC005), Assign Welfare Duty Task (UC007) or Replace Driver (UC008) is executed.
Post Conditions	A ranked list of suitable drivers is displayed.
Priority	High
Frequency of Use	High
Normal Course of Events	**Actor** / **System** 1. UC003, UC005, UC007 or UC008 is executed. / 1.1 System keeps only active drivers who hold a valid licence of the class required for the vehicle. 1.2 System removes drivers who already have a task during the task period or have an approved schedule change on that date. 1.3 System counts each driver's tasks this month, including request tasks, internal tasks and welfare duty, and their weekend tasks. 1.4 System calculates a weighted score for each driver from the number of tasks this month, the number of weekend tasks and years of experience based on the start date of service. 1.5 System displays the drivers ranked by score, with each driver's task counts.
Alternative Courses	Not applicable
Exceptions	1.2 No driver passes the licence and availability checks. The system displays a message that no suitable driver is available.
Includes/Extends	Not applicable
Notes and Issues	• This use case is included in UC003, UC005, UC007 and UC008. • The number of tasks this month has the highest weight, so drivers with fewer tasks are ranked higher. • The licence classes are Class D for cars and vans, GDL for lorries and PSV for buses. • A driver whose licence has expired is not recommended. • The recommendation supports the decision. The Administrative Assistant makes the final choice.

UC010 View Driver Schedule

Use Case ID	UC010
Use Case Name	View Driver Schedule
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant and Head of Admin to view the schedule of all drivers, including request tasks, internal tasks and welfare duty, with the vehicle used for each task.
Preconditions	The Administrative Assistant or Head of Admin is logged in.
Post Conditions	The schedule is displayed.
Priority	High
Frequency of Use	High

Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects Schedule. / 1.1 System displays the schedule of all drivers for the current week, including request tasks, internal tasks and welfare duty, with the vehicle for each task. 2. Administrative Assistant filters by date, driver or vehicle. / 2.1 System displays the matching schedule. 3. Administrative Assistant selects a task. / 3.1 System displays the task details.
Alternative Courses	A1 Monthly view. At step 1, the Administrative Assistant selects Month. The system displays the schedule for the whole month instead of the current week.
Exceptions	2.1 No task matches the filter. The system displays a message that no record is found.
Includes/Extends	Not applicable
Notes and Issues	• The schedule replaces the monthly driver schedule spreadsheet. • Drivers view their own tasks through View Assigned Task (UC023).

UC011 View Report

Use Case ID	UC011
Use Case Name	View Report
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant and Head of Admin to view monthly reports on driver workload, driver replacements and vehicle usage.
Preconditions	The Administrative Assistant or Head of Admin is logged in.
Post Conditions	The report is displayed.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	**Actor** / **System** 1. Administrative Assistant selects Report. / 1.1 System displays the report types, which are Driver Workload, Driver Replacement and Vehicle Usage. 2. Administrative Assistant selects a report type and the month. / 2.1 System displays the report for that month. The Driver Workload report shows the number of request tasks, internal tasks, welfare duties and weekend tasks for each driver. The Driver Replacement report shows each replacement with the date, task, previous driver, new driver and reason. The Vehicle Usage report shows the number of trips and total mileage for each vehicle.
Alternative Courses	Not applicable
Exceptions	2.1 No record is found for the selected month. The system displays a message that no record is found.
Includes/Extends	<<extend>> UC012 Export Report
Notes and Issues	The reports replace the manual counting of tasks in the spreadsheet.

UC012 Export Report

Use Case ID	UC012
Use Case Name	Export Report
Actor	Administrative Assistant, Head of Admin
Description	Allows the Administrative Assistant and Head of Admin to export a displayed report as a PDF or Excel file.
Preconditions	View Report (UC011) is executed and a report is displayed.
Post Conditions	The report file is downloaded.
Priority	Low
Frequency of Use	Low
Normal Course of Events	

Actor	System
1. UC011 is executed and the Administrative Assistant selects Export.	1.1 System displays the file format options, which are PDF and Excel.
2. Administrative Assistant selects PDF or Excel.	2.1 System generates the report file. 2.2 System downloads the file.

Alternative Courses	Not applicable
Exceptions	2.1 The report file cannot be generated. The system displays a message to try again later.
Includes/Extends	Not applicable
Notes and Issues	This use case extends UC011 View Report.

UC013 Register Account

Use Case ID	UC013
Use Case Name	Register Account
Actor	Staff, Head of Department
Description	Allows a first-time user to register an account after verifying the staff ID and UIS email with a verification code.
Preconditions	The user has a UIS staff ID and UIS email and has not registered before.
Post Conditions	The account is created with the selected role.
Priority	High
Frequency of Use	Low
Normal Course of Events	

Actor	System
1. User selects First Time User? Register Now.	1.1 System displays the verification form.
2. User enters the staff ID and UIS email, then selects Request Code.	2.1 System checks that the staff ID and email are not registered. 2.2 System sends a verification code to the email (UC014).
3. User enters the verification code, then selects Submit.	3.1 System checks the code and displays the registration form.
4. User enters the name and phone number, selects the department and role, enters the password and confirm password, then selects Register.	4.1 System creates the account with the selected role. 4.2 System displays the login page with a success message.

Alternative Courses	Not applicable

Exceptions	2.1 The staff ID or email is already registered. The system displays a message that the account already exists. 2.2 The email is not a UIS email. The system displays a message that only UIS email is accepted. 3.1 The verification code is wrong. The system displays a message that verification failed and asks the user to try again. 3.2 The code has expired or was not received. The user selects Resend Code and the system sends a new verification code. 4.1 The passwords do not match. The system displays an error message. 4.2 The user selects Head of Department for a department that already has one. The system displays a message that the department already has a Head of Department.
Includes/Extends	<<include>> UC014 Send Verification Email
Notes and Issues	• The role options are Staff and Head of Department. A user who does not change the role is registered as Staff. • Driver accounts are created by the Administrative Assistant in Create Driver Account (UC001.1). The Administrative Assistant and Head of Admin accounts are created during the initial setup of the system.

UC014 Send Verification Email

Use Case ID	UC014
Use Case Name	Send Verification Email
Actor	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin
Description	Sends a verification code to the user's UIS email so that the user can register an account or reset the password.
Preconditions	Register Account (UC013) or Reset Password (UC015) is executed.
Post Conditions	The verification code is sent to the user's UIS email.
Priority	High
Frequency of Use	Low
Normal Course of Events	**Actor** | **System** 1. UC013 or UC015 is executed. | 1.1 System generates a verification code with an expiry time. 1.2 System sends the code to the user's UIS email. 1.3 System displays a field to enter the code.
Alternative Courses	Not applicable
Exceptions	1.2 The email cannot be sent. The system displays a message to try again later.
Includes/Extends	Not applicable
Notes and Issues	• This use case is included in UC013 Register Account and UC015 Reset Password. • A new code replaces any earlier code for the same user.

UC015 Reset Password

Use Case ID	UC015
Use Case Name	Reset Password
Actor	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin
Description	Allows a user who forgot their password to set a new password using a verification code sent to their UIS email.
Preconditions	The user has a registered and active account.

Post Conditions	The password is changed.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	**Actor** / **System** 1. User selects Forgot Password? on the login page. / 1.1 System displays the reset password form. 2. User enters the staff ID and UIS email, then selects Request Code. / 2.1 System checks that the staff ID and email match a registered account. 2.2 System sends a verification code to the email (UC014). 3. User enters the verification code, then selects Submit. / 3.1 System checks the code and displays the new password form. 4. User enters the new password and confirm password, then selects Save. / 4.1 System updates the password. 4.2 System displays the login page with a success message.
Alternative Courses	Not applicable
Exceptions	2.1 The staff ID and email do not match any active account. The system displays a message that the account is not found. 3.1 The verification code is wrong. The system displays a message that verification failed and asks the user to try again. 3.2 The code has expired or was not received. The user selects Resend Code and the system sends a new verification code. 4.1 The passwords do not match. The system displays an error message.
Includes/Extends	<<include>> UC014 Send Verification Email
Notes and Issues	Login is not a use case. It is a precondition of the other use cases and is covered in Section 4.2.

UC016 Manage Profile

Use Case ID	UC016
Use Case Name	Manage Profile
Actor	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin
Description	Allows a user to manage their own profile. This use case covers view profile, update profile and change password, which are described in UC016.1, UC016.2 and UC016.3.
Preconditions	The user is logged in.
Post Conditions	The profile is displayed or updated, or the password is changed.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	Not applicable
Alternative Courses	Not applicable
Exceptions	Not applicable
Includes/Extends	Not applicable
Notes and Issues	• Users cannot change their own staff ID, UIS email, department or role. • A driver's licence details are updated by the Administrative Assistant in Update Staff and Driver Account (UC001.3).

UC016.1 View Profile

Use Case ID	UC016.1
Use Case Name	View Profile
Actor	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin
Description	Allows a user to view their own profile details.
Preconditions	The user is logged in.
Post Conditions	The profile is displayed.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	**Actor** / **System** 1. User selects Profile. / 1.1 System displays the staff ID, name, UIS email, phone number, address, department and role. 1.2 If the user is a driver, the system also displays the licence class, licence expiry date and start date of service.
Alternative Courses	Not applicable
Exceptions	Not applicable
Includes/Extends	Not applicable
Notes and Issues	Not applicable

UC016.2 Update Profile

Use Case ID	UC016.2
Use Case Name	Update Profile
Actor	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin
Description	Allows a user to update their own name, phone number and address.
Preconditions	The user is logged in.
Post Conditions	The profile is updated.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	**Actor** / **System** 1. User selects Profile. / 1.1 System displays the profile details. 2. User selects Edit. / 2.1 System displays the profile form with the current name, phone number and address. 3. User updates the details, then selects Save. / 3.1 System checks the details. 3.2 System updates the profile and displays a success message.
Alternative Courses	Not applicable
Exceptions	3.1 A required field is empty or in the wrong format. The system displays a message to fill in all required fields correctly.
Includes/Extends	Not applicable
Notes and Issues	Not applicable

UC016.3 Change Password

Use Case ID	UC016.3
Use Case Name	Change Password
Actor	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin
Description	Allows a user to change their own password while logged in.
Preconditions	The user is logged in.
Post Conditions	The password is changed.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	**Actor** / **System** 1. User selects Profile. / 1.1 System displays the profile details. 2. User selects Change Password. / 2.1 System displays the change password form. 3. User enters the current password, new password and confirm password, then selects Save. / 3.1 System checks the current password and that the new passwords match. 3.2 System updates the password and displays a success message.
Alternative Courses	Not applicable
Exceptions	3.1 The current password is wrong or the new passwords do not match. The system displays an error message.
Includes/Extends	Not applicable
Notes and Issues	A driver can use this use case to change the default password given in Create Driver Account (UC001.1).

UC017 View Request

Use Case ID	UC017
Use Case Name	View Request
Actor	Staff, Head of Department, Driver, Administrative Assistant, Head of Admin
Description	Allows a user to view the list of requests and the details of each request, including its current status and status history.
Preconditions	The user is logged in.
Post Conditions	The request details are displayed.
Priority	High
Frequency of Use	High

<table>
<tr><td>Normal Course of Events</td><td><table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. User selects My Request.</td><td>1.1 System displays the user's own requests with their request ID, vehicle type, departure date and status.</td></tr>
<tr><td>2. User filters the list by status or date.</td><td>2.1 System displays the matching requests.</td></tr>
<tr><td>3. User selects a request.</td><td>3.1 System displays the request details, supporting document, assigned driver and vehicle, comments and status history with the reason for each change.</td></tr>
<tr><td>4. If the request is an approved Pay and Claim or Personal request, the user selects Print Approval Letter.</td><td>4.1 System displays the approval letter. For Pay and Claim, the letter shows the request details, approval dates and approvers. For Personal, the letter also shows the distance and the charge.
4.2 System lets the user print or download the letter as a PDF.</td></tr>
</table></td></tr>
<tr><td>Alternative Courses</td><td>A1 Department requests. At step 1, a Head of Department selects Department Request. The system displays the requests from the user's department.
A2 All requests. At step 1, the Administrative Assistant or Head of Admin selects All Request. The system displays the requests from all departments.</td></tr>
<tr><td>Exceptions</td><td>2.1 No request matches the filter. The system displays a message that no record is found.</td></tr>
<tr><td>Includes/Extends</td><td>Not applicable</td></tr>
<tr><td>Notes and Issues</td><td>• The request status is Pending, Approved by Head of Department, Driver Assigned, Approved, Rejected, Cancelled, Lapsed or Completed.
• A lapsed request is shown with its own status, so staff know that it was not cancelled by the Vehicle Section.
• The claim payment and the rental payment are handled by the Bursary outside the system.</td></tr>
</table>

UC018 Submit Vehicle Request

Use Case ID	UC018
Use Case Name	Submit Vehicle Request
Actor	Staff, Head of Department, Driver
Description	Allows a user to submit a vehicle request. The user selects the vehicle type and whether a driver is needed, then fills in the trip details. Lorries can also be requested for personal purposes at a charge per kilometre.
Preconditions	The user is logged in.
Post Conditions	1. The request is saved with the Pending status. 2. The requested vehicles are reserved for the trip period. 3. The approval email is sent.
Priority	High
Frequency of Use	High

<table>
<tr><td>Normal Course of Events</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. User selects New Request.</td><td>1.1 System displays the vehicle types, which are car, MPV, van, bus and lorry.</td></tr>
<tr><td>2. User selects a vehicle type.</td><td>2.1 System displays the request form.</td></tr>
<tr><td>3. User selects the purpose type, enters the purpose, destination with full address, waiting location, departure date and time, return date and time, number of passengers and quantity of vehicles, uploads the supporting letter, then selects Submit.</td><td>3.1 System checks the availability of the vehicles for the trip period.
3.2 If the departure date is less than two days away, the system displays a warning and marks the request as Urgent.
3.3 System saves the request with the Pending status.
3.4 System reserves the requested quantity of the vehicle type for the trip period.
3.5 System sends an approval email to the Head of Department (UC019).</td></tr>
</table>
</td></tr>
<tr><td>Alternative Courses</td><td>A1 Pay and Claim. At step 2, the user selects a car. The system displays With Driver and Pay and Claim. The user selects Pay and Claim and uses their own car. No vehicle is reserved.
A2 Self Drive lorry. At step 2, the user selects a lorry. If the user is a driver with a GDL licence, the system displays With Driver and Self Drive. The user selects Self Drive and drives the lorry without a UIS driver. Self Drive is not shown to other users.
A3 Personal purpose. At step 3, the user selects a lorry and selects Personal as the purpose type. The user enters the round-trip distance instead of uploading a supporting letter. The system displays the estimated charge at RM3 per kilometre and sends the request to the Administrative Assistant instead of the Head of Department.</td></tr>
<tr><td>Exceptions</td><td>3.1 Not enough vehicles of the selected type are available for the trip period. The system displays a message that the vehicles are not available.
3.3 A required field is empty. The system displays a message to fill in all required fields.
3.4 The purpose type is Official or Unofficial and no supporting letter is uploaded. The system displays a message that the supporting letter is required.
3.5 The return date and time is earlier than the departure date and time. The system displays an error message.</td></tr>
<tr><td>Includes/Extends</td><td><<include>> UC019 Send Approval Email; <<extend>> UC020 Send Approval Reminder Email</td></tr>
<tr><td>Notes and Issues</td><td>• The purpose types are Official and Unofficial for all vehicles, and Personal for lorries only. Official and Unofficial require a supporting letter. Personal does not, and the user pays the charge because the lorry is used for personal purposes.
• The vehicle is reserved when the request is submitted, so vehicles are given on a first-submitted, first-served basis. The reservation covers the departure time until one hour after the return time.
• If the Head of Department does not approve the request within two days, the request changes to Lapsed and the reservation is released. The reservation is also released if the request is cancelled or rejected. Pay and Claim requests do not reserve any vehicle.
• Pay and Claim and Self Drive requests go to the Head of Department, then the Head of Admin.
• Personal requests go to the Administrative Assistant, then the Head of Admin.
• A request from a Head of Department is sent to the head of that Head of Department for approval.
• All requests are made in the system, including urgent requests. The paper form is no longer used.</td></tr>
</table>

UC019 Send Approval Email

Use Case ID	UC019
Use Case Name	Send Approval Email
Actor	Staff, Head of Department, Driver

Description	Sends an email to the approver when a new request is submitted, so the approver can act on it.
Preconditions	Submit Vehicle Request (UC018) is executed.
Post Conditions	The approval email is sent to the approver.
Priority	High
Frequency of Use	High
Normal Course of Events	**Actor** / **System** 1. UC018 is executed. / 1.1 System finds the approver for the request. The approver is the Head of Department, the head of that Head of Department for a request made by a Head of Department, or the Administrative Assistant for a Personal request. 1.2 System sends an email with the request details and a link to the request. 1.3 System records the date and time the email was sent.
Alternative Courses	Not applicable
Exceptions	1.2 The email cannot be sent. The system keeps the request and shows it in the approver's pending list. The system tries to send the email again later.
Includes/Extends	Not applicable
Notes and Issues	• This use case is included in UC018 Submit Vehicle Request. • The approver also receives an in-system notification.

UC020 Send Approval Reminder Email

Use Case ID	UC020
Use Case Name	Send Approval Reminder Email
Actor	Head of Department
Description	Sends a reminder email to the Head of Department when a request is still waiting for approval after one day.
Preconditions	1. Submit Vehicle Request (UC018) is executed. 2. The request status is Pending. 3. One day has passed since the request was submitted.
Post Conditions	The reminder email is sent to the Head of Department.
Priority	Medium
Frequency of Use	Medium
Normal Course of Events	**Actor** / **System** 1. UC018 is executed. / 1.1 System checks the pending requests every hour. 1.2 System finds requests that have been pending for one day. 1.3 System sends a reminder email to the Head of Department with the time left before the request lapses.
Alternative Courses	Not applicable
Exceptions	1.3 The email cannot be sent. The system tries to send the email again at the next check.
Includes/Extends	Not applicable

Notes and Issues	• This use case extends UC018 Submit Vehicle Request. • The reminder is sent once for each request. If the request is still not approved after two days, it changes to Lapsed.

UC021 Cancel Vehicle Request

Use Case ID	UC021
Use Case Name	Cancel Vehicle Request
Actor	Staff, Head of Department, Driver
Description	Allows a user to cancel their own vehicle request before the trip. Also allows the Head of Department to cancel a vehicle request from their department. A reason is required for every cancellation.
Preconditions	1. The user is logged in. 2. The request has not started and is not Rejected, Cancelled, Lapsed or Completed.
Post Conditions	1. The request status is changed to Cancelled. 2. The vehicle reservation is released.
Priority	High
Frequency of Use	Medium
Normal Course of Events	**Actor** / **System** 1. User selects My Request, then selects a request. / 1.1 System displays the request details. 2. User selects Cancel Request. / 2.1 System displays a pop-up form to enter the reason for cancellation. 3. User enters the reason, then selects Confirm. / 3.1 System changes the status to Cancelled and records the reason. 3.2 System releases the vehicle reservation. 3.3 System sends a notification to the related users.
Alternative Courses	A1 Head of Department. At step 1, a Head of Department selects Department Request, then selects a request from their department. The flow continues from step 2.
Exceptions	3.1 The reason is empty. The system displays a message that the reason is required.
Includes/Extends	Not applicable
Notes and Issues	• The related users are the requester, the Head of Department, the Administrative Assistant and the assigned driver, if any. • The reason is shown in the status history and in the notification, so the requester knows who cancelled the request and why.

UC022 Request Schedule Change

Use Case ID	UC022
Use Case Name	Request Schedule Change
Actor	Driver
Description	Allows a driver to inform the Vehicle Section of dates when they cannot work, such as a weekend, or to ask to be removed from a task assigned to them.
Preconditions	The driver is logged in.
Post Conditions	The schedule change request is sent to the Head of Admin with the Pending status.
Priority	Medium

Frequency of Use	Medium
Normal Course of Events	**Actor** / **System** 1. Driver selects Schedule Change, then selects New Request. / 1.1 System displays the schedule change form. 2. Driver selects the unavailable date or dates, enters the reason, then selects Submit. / 2.1 System checks the details. 2.2 System saves the request with the Pending status. 2.3 System sends a notification to the Head of Admin.
Alternative Courses	A1 Remove from a task. At step 1, the driver selects My Task, selects the task, then selects Request Change. The system displays the form with the task details. The driver enters the reason, then selects Submit. The system saves the request and sends a notification to the Head of Admin.
Exceptions	2.1 The reason is empty or no date is selected. The system displays a message to fill in all required fields. 2.1 The selected date has already passed. The system displays an error message.
Includes/Extends	Not applicable
Notes and Issues	• Schedule changes are approved by the Head of Admin in UC027. • Drivers with an approved schedule change are not recommended for tasks on those dates. • The driver can view the status of their schedule change requests in the Schedule Change list and cancel a request that is still Pending.

UC023 View Assigned Task

Use Case ID	UC023
Use Case Name	View Assigned Task
Actor	Driver
Description	Allows a driver to view the tasks assigned to them, including request tasks, internal tasks and welfare duty.
Preconditions	The driver is logged in.
Post Conditions	The assigned tasks are displayed.
Priority	High
Frequency of Use	High
Normal Course of Events	**Actor** / **System** 1. Driver selects My Task. / 1.1 System displays the driver's upcoming approved tasks with their date, time, task type and vehicle. 2. Driver selects a task. / 2.1 System displays the date, time, vehicle, destination with full address, waiting location, purpose, officer name and officer phone number.
Alternative Courses	A1 Past tasks. At step 1, the driver selects Past Tasks. The system displays the driver's completed tasks.
Exceptions	1.1 The driver has no upcoming task. The system displays a message that no task is assigned.
Includes/Extends	Not applicable
Notes and Issues	The task details replace the WhatsApp messages typed by hand.

UC024 Manage Assigned Task Report

Use Case ID	UC024
Use Case Name	Manage Assigned Task Report
Actor	Driver
Description	Allows a driver to manage the report of a completed task. The report records the trip details and updates the vehicle's mileage, fuel level and status. This use case covers create, view and update task report, which are described in UC024.1, UC024.2 and UC024.3.
Preconditions	1. The driver is logged in. 2. The task has been approved and assigned to the driver.
Post Conditions	The task report is created, displayed or updated.
Priority	High
Frequency of Use	High
Normal Course of Events	Not applicable
Alternative Courses	Not applicable
Exceptions	Not applicable
Includes/Extends	Not applicable
Notes and Issues	• The report applies to request tasks, internal tasks and welfare duty. • A task report cannot be deleted because it updates the vehicle's records.

UC024.1 Create Task Report

Use Case ID	UC024.1
Use Case Name	Create Task Report
Actor	Driver
Description	Allows a driver to create a report after finishing a task. The report updates the vehicle's mileage, fuel level and status.
Preconditions	1. The driver is logged in. 2. The task has been approved and assigned to the driver. 3. The departure time of the task has passed.
Post Conditions	1. The report is saved and the task status is changed to Completed. 2. The vehicle's mileage, fuel level and status are updated.
Priority	High
Frequency of Use	High

<table>
<tr><td>Normal Course of Events</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Driver selects My Task, then selects a task.</td><td>1.1 System displays the task details.</td></tr>
<tr><td>2. Driver selects Create Report.</td><td>2.1 System displays the task report form with the current mileage of the vehicle.</td></tr>
<tr><td>3. Driver enters the actual departure time, actual return time, end mileage and fuel level, then selects the vehicle condition.</td><td>3.1 If the vehicle condition is Problem, the system displays the vehicle problem form.</td></tr>
<tr><td>4. If the vehicle has a problem, the driver selects the problem type, enters the description, uploads a photo and states whether the vehicle can still be used. The driver then selects Submit.</td><td>4.1 System checks the details.
4.2 System saves the report and changes the task status to Completed.
4.3 System updates the vehicle's current mileage and fuel level. If the vehicle cannot be used, the system changes the vehicle status to Under Maintenance.
4.4 System sends a notification to the Administrative Assistant and highlights any vehicle problem.</td></tr>
</table>
</td></tr>
<tr><td>Alternative Courses</td><td>Not applicable</td></tr>
<tr><td>Exceptions</td><td>1.1 The task has not reached its departure time. The system does not show the Create Report button.
4.1 A required field is empty. The system displays a message to fill in all required fields.
4.1 The end mileage is lower than the current mileage. The system displays a message that the mileage is not valid.</td></tr>
<tr><td>Includes/Extends</td><td>Not applicable</td></tr>
<tr><td>Notes and Issues</td><td>The problem types are Plate Number, Tyre, Lights, Body Damage, Engine and Others. If the vehicle can still be used, its status stays Active.</td></tr>
</table>

UC024.2 View Task Report

<table>
<tr><td>Use Case ID</td><td>UC024.2</td></tr>
<tr><td>Use Case Name</td><td>View Task Report</td></tr>
<tr><td>Actor</td><td>Driver</td></tr>
<tr><td>Description</td><td>Allows a driver to view the report of a past task.</td></tr>
<tr><td>Preconditions</td><td>1. The driver is logged in.
2. The driver has at least one completed task with a report.</td></tr>
<tr><td>Post Conditions</td><td>The task report is displayed.</td></tr>
<tr><td>Priority</td><td>Medium</td></tr>
<tr><td>Frequency of Use</td><td>Medium</td></tr>
<tr><td>Normal Course of Events</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Driver selects My Task, then selects Past Tasks.</td><td>1.1 System displays the driver's completed tasks.</td></tr>
<tr><td>2. Driver selects a task.</td><td>2.1 System displays the task report with the actual times, end mileage, fuel level, vehicle condition and any vehicle problem.</td></tr>
</table>
</td></tr>
<tr><td>Alternative Courses</td><td>Not applicable</td></tr>
<tr><td>Exceptions</td><td>1.1 The driver has no completed task. The system displays a message that no record is found.</td></tr>
<tr><td>Includes/Extends</td><td>Not applicable</td></tr>
</table>

Notes and Issues	Not applicable

UC024.3 Update Task Report

Use Case ID	UC024.3
Use Case Name	Update Task Report
Actor	Driver
Description	Allows a driver to correct the report of a past task.
Preconditions	1. The driver is logged in. 2. The task report exists.
Post Conditions	The report and the vehicle details are updated.
Priority	Medium
Frequency of Use	Low
Normal Course of Events	**Actor** / **System** 1. Driver selects My Task, then selects Past Tasks. / 1.1 System displays the driver's completed tasks. 2. Driver selects a task report, then selects Edit. / 2.1 System displays the report form with the current details. 3. Driver updates the details, then selects Save. / 3.1 System checks the details. 3.2 System updates the report and the vehicle details. 3.3 System sends a notification to the Administrative Assistant and displays a success message.
Alternative Courses	Not applicable
Exceptions	3.1 A required field is empty. The system displays a message to fill in all required fields. 3.1 The end mileage is lower than the vehicle's mileage before the trip. The system displays a message that the mileage is not valid.
Includes/Extends	Not applicable
Notes and Issues	Not applicable

UC025 Approve/Reject Vehicle Request

Use Case ID	UC025
Use Case Name	Approve/Reject Vehicle Request
Actor	Head of Department, Head of Admin
Description	Allows the Head of Department to approve or reject a vehicle request from their department. Also allows the Head of Admin to approve or reject Pay and Claim and Self Drive requests after the Head of Department approves them.
Preconditions	1. The user is logged in. 2. The vehicle request is waiting for the user's approval.
Post Conditions	The vehicle request status is updated.
Priority	High
Frequency of Use	High

<table>
<tr><td>Normal Course of Events</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Head of Department selects Department Request.</td><td>1.1 System displays the pending requests from the department, with Urgent requests at the top.</td></tr>
<tr><td>2. Head of Department selects a request.</td><td>2.1 System displays the request details and supporting letter.</td></tr>
<tr><td>3. Head of Department selects Approve.</td><td>3.1 System changes the status to Approved by Head of Department.
3.2 System sends the request to the Administrative Assistant for driver assignment, or to the Head of Admin if the request is Pay and Claim or Self Drive.
3.3 System sends a notification to the requester.</td></tr>
</table>
</td></tr>
<tr><td>Alternative Courses</td><td>A1 Reject. At step 3, the Head of Department selects Reject. The system displays the comment box (UC028). The Head of Department enters the reason, then selects Confirm. The system changes the status to Rejected, releases the vehicle reservation and sends a notification to the requester with the reason.
A2 Head of Admin approval. At step 1, the Head of Admin selects Pending Approval, selects a Pay and Claim or Self Drive request, then selects Approve or Reject. The system updates the status and sends a notification to the requester.</td></tr>
<tr><td>Exceptions</td><td>A1.1 The reason is empty when rejecting. The system displays a message that the reason is required.</td></tr>
<tr><td>Includes/Extends</td><td><<extend>> UC028 Add Comment (extension point Reject)</td></tr>
<tr><td>Notes and Issues</td><td>• No comment is needed when a request is approved.
• A request that is not approved by the Head of Department within two days changes to Lapsed, and the vehicle reservation is released.
• Requests with a driver are approved by the Head of Admin in UC026, after the Administrative Assistant assigns the driver.</td></tr>
</table>

UC026 Approve/Reject Driver Assignment

Use Case ID	UC026
Use Case Name	Approve/Reject Driver Assignment
Actor	Head of Admin
Description	Allows the Head of Admin to approve or reject the vehicle and driver assigned by the Administrative Assistant for request tasks, internal tasks, welfare duty and driver replacements.
Preconditions	1. The Head of Admin is logged in. 2. The task status is Driver Assigned.
Post Conditions	The task status is updated.
Priority	High
Frequency of Use	High

Normal Course of Events	**Actor** / **System** 1. Head of Admin selects Pending Approval. / 1.1 System displays the tasks with assigned drivers, with Urgent requests at the top. 2. Head of Admin selects a task. / 2.1 System displays the task details, assigned vehicles, assigned drivers and each driver's task count this month. 3. Head of Admin selects Approve. / 3.1 System changes the status to Approved. 3.2 System sends a notification to the requester, the assigned drivers and the Administrative Assistant. For a driver replacement, the system also notifies the previous driver.
Alternative Courses	A1 Reject. At step 3, the Head of Admin selects Reject. The system displays the comment box (UC028). The Head of Admin enters the reason, then selects Confirm. The system returns the task to the Administrative Assistant with the reason, so a new driver can be assigned. The vehicle reservation is kept.
Exceptions	A1.1 The reason is empty when rejecting. The system displays a message that the reason is required.
Includes/Extends	<<extend>> UC028 Add Comment (extension point Reject)
Notes and Issues	• Rejecting a driver assignment does not reject the request. It only returns the task to the Administrative Assistant. • The Head of Admin knows which drivers are on leave, so this approval checks that the assigned drivers are available. • This use case also approves driver replacements made in UC008.

UC027 Approve/Reject Schedule Change

Use Case ID	UC027
Use Case Name	Approve/Reject Schedule Change
Actor	Head of Admin
Description	Allows the Head of Admin to approve or reject a schedule change requested by a driver.
Preconditions	1. The Head of Admin is logged in. 2. The schedule change request status is Pending.
Post Conditions	The schedule change request status is updated.
Priority	Medium
Frequency of Use	Medium
Normal Course of Events	**Actor** / **System** 1. Head of Admin selects Schedule Change. / 1.1 System displays the pending schedule change requests. 2. Head of Admin selects a request. / 2.1 System displays the driver, dates, reason and any tasks the driver has on those dates. 3. Head of Admin selects Approve. / 3.1 System changes the status to Approved. 3.2 System marks the driver as unavailable on those dates. 3.3 If the driver has a task on those dates, the system sends the task to the Administrative Assistant to replace the driver (UC008). 3.4 System sends a notification to the driver.

Alternative Courses	A1 Reject. At step 3, the Head of Admin selects Reject. The system displays the comment box (UC028). The Head of Admin enters the reason, then selects Confirm. The system changes the status to Rejected and sends a notification to the driver with the reason.
Exceptions	A1.1 The reason is empty when rejecting. The system displays a message that the reason is required.
Includes/Extends	<<extend>> UC028 Add Comment (extension point Reject)
Notes and Issues	A driver marked as unavailable is not recommended for tasks on those dates.

UC028 Add Comment

Use Case ID	UC028
Use Case Name	Add Comment
Actor	Head of Department, Head of Admin
Description	Allows the approver to enter the reason when rejecting a vehicle request, a driver assignment or a schedule change.
Preconditions	Approve/Reject Vehicle Request (UC025), Approve/Reject Driver Assignment (UC026) or Approve/Reject Schedule Change (UC027) is executed and the approver selects Reject.
Post Conditions	The reason is saved and shown to the affected users.
Priority	High
Frequency of Use	Medium
Normal Course of Events	**Actor** / **System** 1. UC025, UC026 or UC027 is executed and the approver selects Reject. / 1.1 System displays the comment box. 2. Approver enters the reason, then selects Confirm. / 2.1 System saves the reason with the date, time and approver name. 2.2 System returns to the use case that called it.
Alternative Courses	Not applicable
Exceptions	2.1 The reason is empty. The system displays a message that the reason is required.
Includes/Extends	Not applicable
Notes and Issues	• This use case extends UC025, UC026 and UC027. It is used only when rejecting. • Cancel Vehicle Request (UC021) uses its own pop-up form for the reason and does not use this use case.